

Adı - Soyadı: İzzet TEKBAŞ

16/06/2020

No: 18058056

Analiz-II

$$1) \int \frac{2}{x \cdot \sqrt{x^2-1}} dx = \int \frac{2 dx}{x^2 \sqrt{1 - (\frac{1}{x})^2}} = 2 \int \frac{-\cancel{\cos t} dt}{\sqrt{1 - \cancel{\sin^2 t}}} = -2 \int dt = -2t + C$$
$$= -2 \arcsin \frac{1}{x} + C$$

$$\frac{1}{x} = \sin t$$

$$-\frac{dx}{x^2} = \cos t dt$$

$$\int_1^{\infty} \frac{2}{x \sqrt{x^2-1}} dx = \lim_{b \rightarrow \infty} \int_1^b \frac{2}{x \sqrt{x^2-1}} dx = \lim_{b \rightarrow \infty} \left(-2 \arcsin \frac{1}{x} \right) \Big|_1^b$$

$$= \lim_{b \rightarrow \infty} \left(-2 \arcsin \left(\frac{1}{b} \right) + 2 \arcsin 1 \right) = -2 \cdot 0 + 2 \cdot \frac{\pi}{2}$$
$$= \pi //$$

Answer: π

İ. Moller

Name - Surname: İzzet TEKBAŞ

16/06/2020

No: 18058056

Analiz - II

$$2) \sum_{n=1}^{\infty} \frac{(x+3)^n}{(5n-1)5^n}$$

According to D' Alembert Ratio Test;

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1 \quad \text{if } a_n \text{ is convergent}$$

$$\left(\lim_{n \rightarrow \infty} \frac{1 \cdot (1 + \frac{4}{5})}{1 \cdot (5 - \frac{1}{5})} = \frac{5}{5} = 1 \right)$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{(x+3)^{n+1}}{(5n+4)5^{n+1}} \cdot \frac{(5n-1)5^n}{(x+3)^n} \right| = \frac{|x+3|}{5} \cdot \lim_{n \rightarrow \infty} \frac{5n+4}{5n-1}$$

$$= \frac{|x+3|}{5} < 1 \Rightarrow |x+3| < 5$$

$$\Rightarrow R=5 \text{ and } -8 < x < 2$$

Btw, we can find R with formula $R = \frac{1}{\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|}$, where $c_n = \frac{1}{(5n-1)5^n}$

Let check convergence at endpoints;

$$i) \text{ for } x=2, \sum_{n=1}^{\infty} \frac{(x+3)^n}{(5n-1)5^n} = \sum_{n=1}^{\infty} \frac{5^n}{(5n-1)5^n} = \sum_{n=1}^{\infty} \frac{1}{5n-1} > \sum_{n=1}^{\infty} \frac{1}{5n} = \frac{1}{5} \sum_{n=1}^{\infty} \frac{1}{n}$$

$\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent according to p -test ($p=1 \nless 1$)

$\Rightarrow \frac{1}{5} \sum_{n=1}^{\infty} \frac{1}{n}$ is divergent. Then $\sum_{n=1}^{\infty} \frac{1}{5n-1}$ is divergent according to comparison test

Thus, the series is divergent for $x=2$.

$$ii) \text{ for } x=-8; \sum_{n=1}^{\infty} \frac{(x+3)^n}{(5n-1)5^n} = \sum_{n=1}^{\infty} (-1)^n \frac{1}{5n-1}, \text{ if } a_n = \frac{1}{5n-1} \Rightarrow \begin{cases} i) a_n > 0 \\ ii) a_{n+1} = \frac{1}{5n+4} < a_n = \frac{1}{5n-1} \end{cases}$$

Since the conditions of alternating test is satisfied, the series is convergent at $x=-8$. It is also conditional convergence, because there is no convergence at $x=2$

(a_n is decreasing)

$$iii) \lim_{n \rightarrow \infty} a_n = 0$$

Answer: $[-8, 2)$

i. values

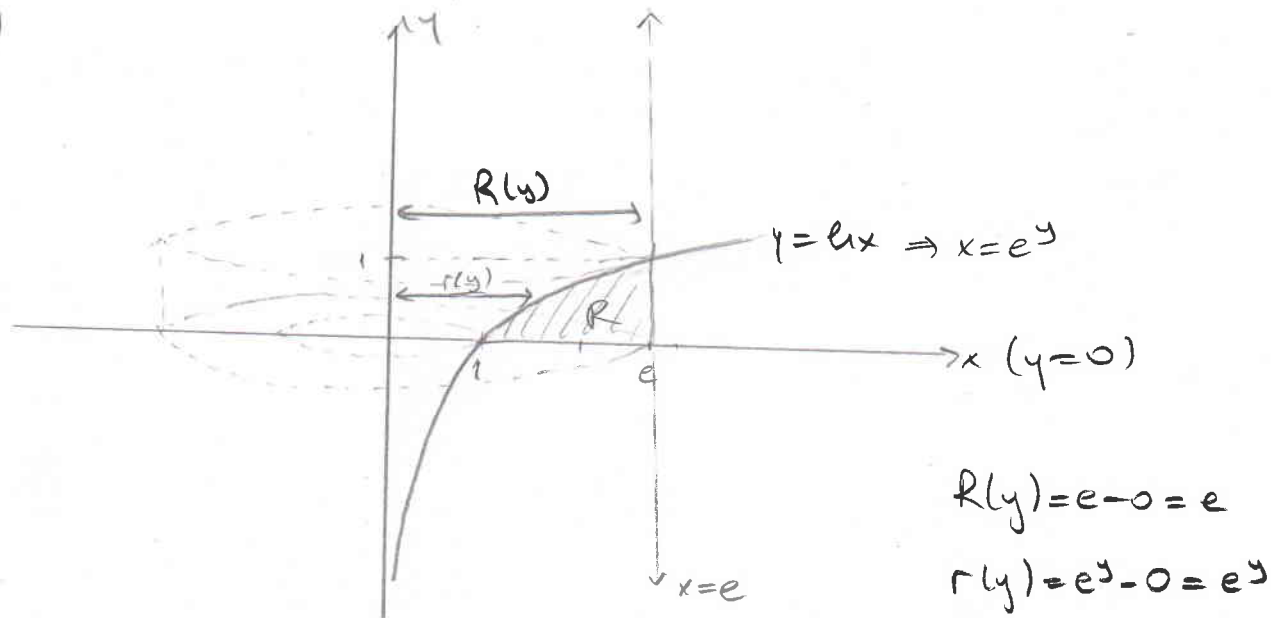
Name - Surname: İzzet TEKBAŞ

16/06/2020

No: 18058056

Analiz-II

3)



$$V = \pi \int_0^1 [R^2(y) - r^2(y)] dy = \pi \int_0^1 (e^2 - (e^y)^2) dy$$

$$= \pi \int_0^1 e^2 dy - \pi \int_0^1 e^{2y} dy = e^2 \pi y \Big|_0^1 - \frac{\pi e^{2y}}{2} \Big|_0^1$$

$$= e^2 \pi - \frac{\pi e^2}{2} + \frac{\pi}{2}$$

$$= \frac{\pi}{2} (1 + e^2)$$

$$\text{Answers a: } \pi \int_0^1 (e^2 - e^{2y}) dy$$

$$\text{Answers b: } \frac{\pi}{2} (1 + e^2)$$

J. M. Elles